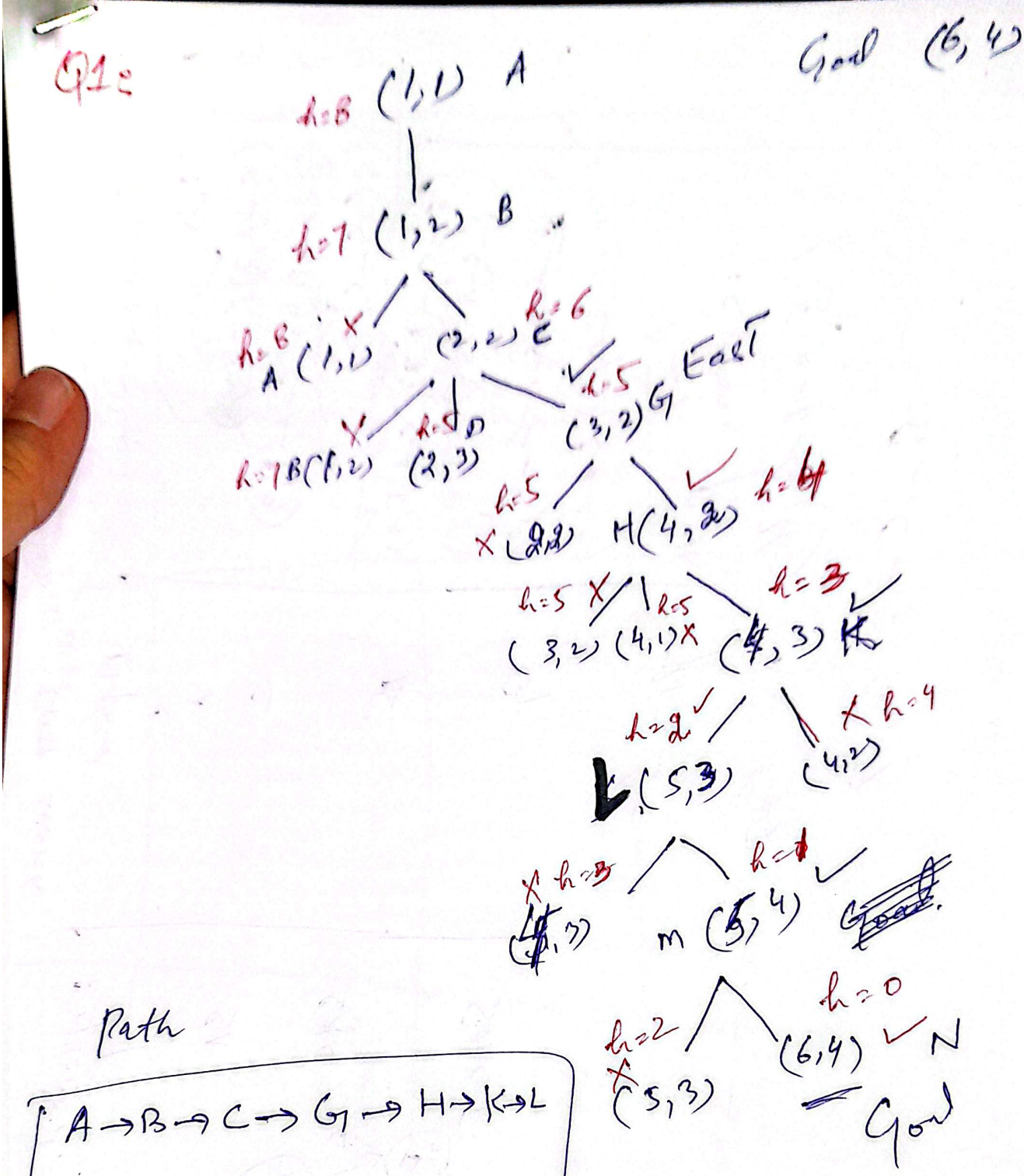
Q1:

Goal (6, 4)



Path

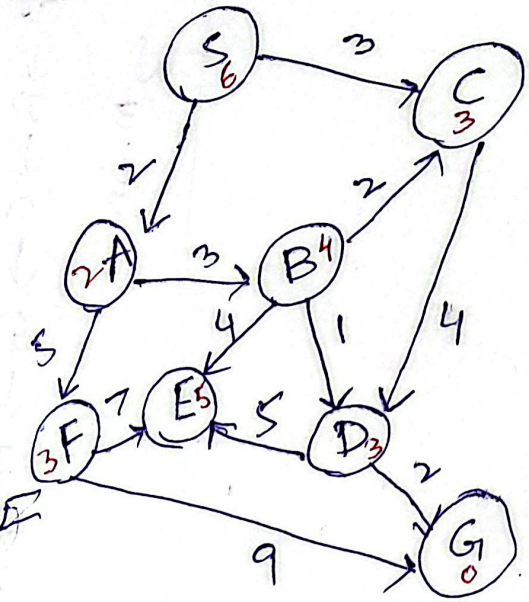
A → B → C → G → H → K → L

→ M → N

Q2 (a)

Best - First Search

Steps	Frontier	Expand	Explored
1	(S, 6)	S	ϕ
2	(S-A, 2) (S→C, 3)	A	S
3	(S-A-B, 4) (S-A-F, 3) (S-C, 3)	C	SA
4	(S-C-D, 3) (S-A-F, 3) (S-A-B, 4)	F	SAC
	~~S-C-F~~ (S-A-F-G, 0) (S-C-D, 3) (S-A-B, 4)	G	SACF



is this path optimal?

(S-A-F-G, 16) no cost time is another path

S-C-D-G, 9

Q2 (b)

UCS

step	Frontier	Expand	Explored list
1	(S, 0)	S	{ }
2	(S-A, 2) (S-C, 3)	A	S
3	(S-A-B, 5) (S-A-F, 7) (S-C, 3)	C	SA
4	(S-A-B, 5) (S-A-F, 7) (S-C-D, 7)	B	SAC
	~~(S-A-B-C, 7)~~ (S-A-B-E, 9) (S-A-B-D, 6) (S-A-F, 7) ~~(S-C-D 7)~~	D	SACB
6	(S-A-B-D-G, 8) ~~(S-A-B-D-E, 11)~~ (S-A-B-E, 9) (S-A-F, 7)	F	SACBD
7	~~(S-A-F-G, 16)~~ (S-A-F-E, 14) (S-A-B-E, 9) (S-A-B-D-G, 8)	G	SACBDF

Path (S-A-B-D-G, 8)

yes optimal.

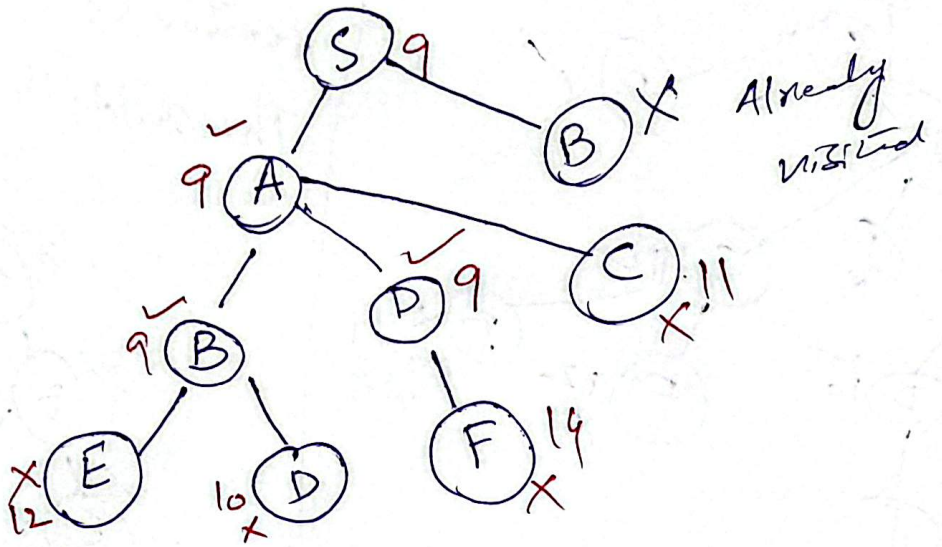
Q4: IDA*

iteration 1 :

Threshold: 9

Pruned list:
12, 10, 14, 11

visited:
SABD

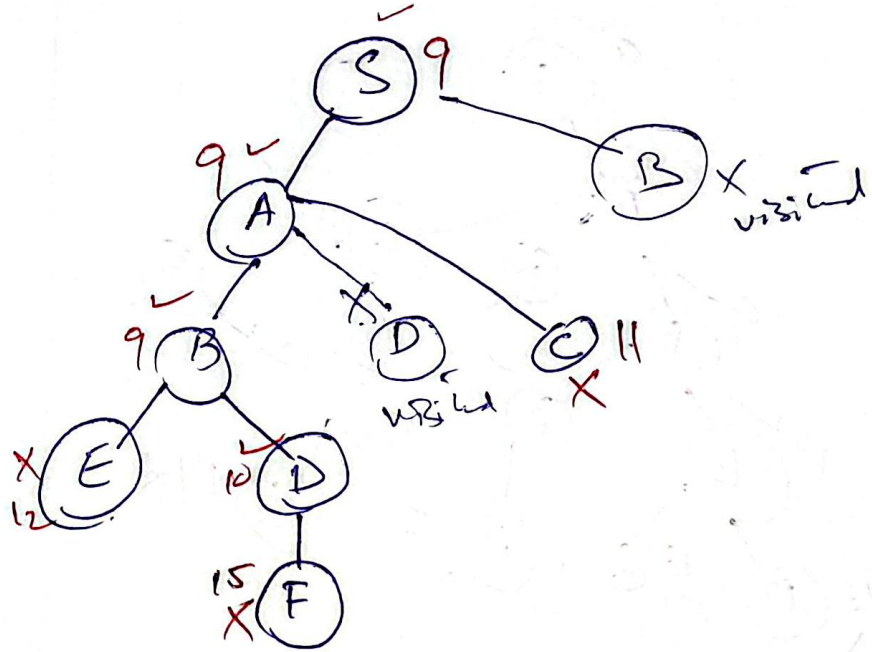


iteration 2 :

Threshold : 10

Pruned List:
12, 15, 11

visited list:
SABD

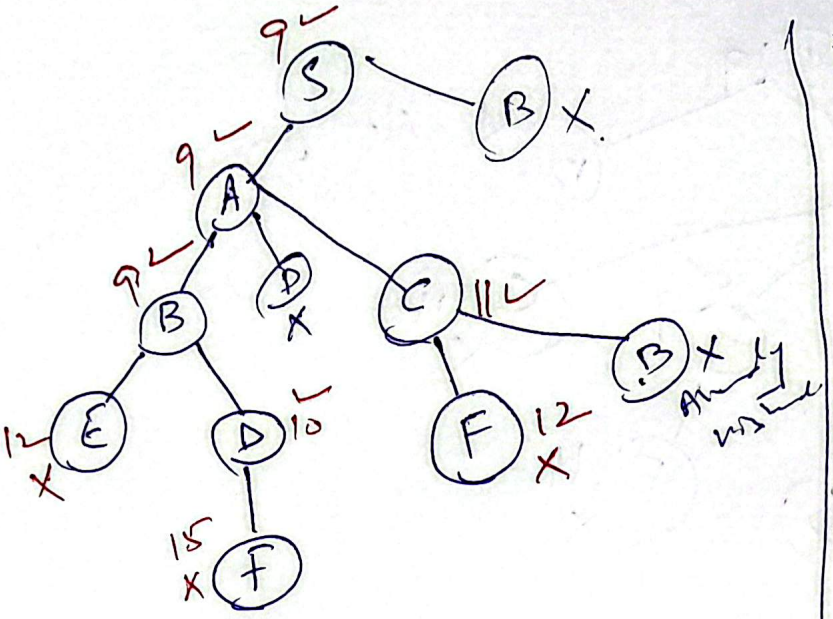


Iteration 3:

Threshold: 11

Pruned list:
12, 15,

visited list
SABDC

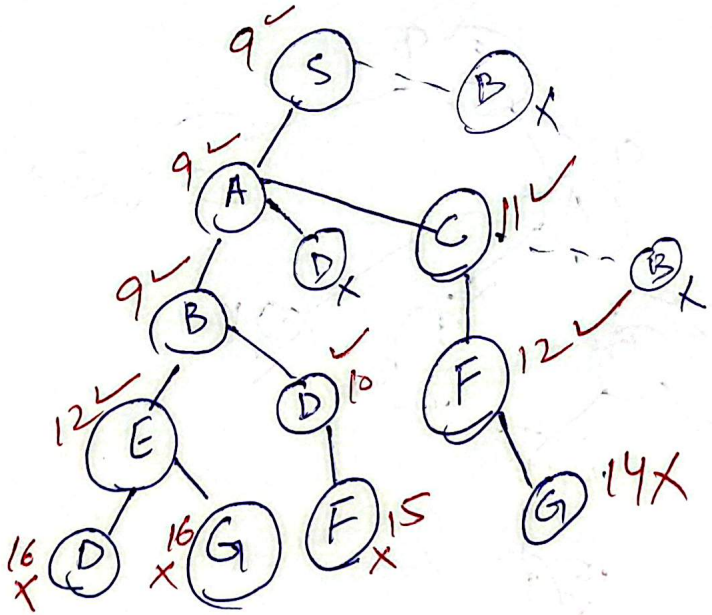


Iteration 4:

Threshold: 12

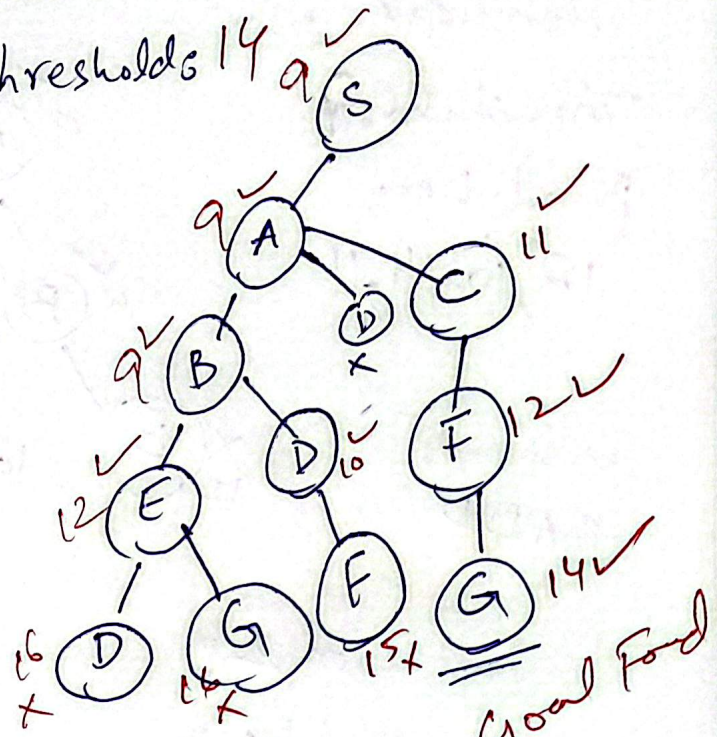
Pruned list:
16, 15, 14

visited list:
SABEDCF



Iteration 5:

Threshold 14



Pruned list:
16, 15

visited list:
SABEDCFG

Q5: MinMAX

